Relationships between two variables

REFERENCE SHEET

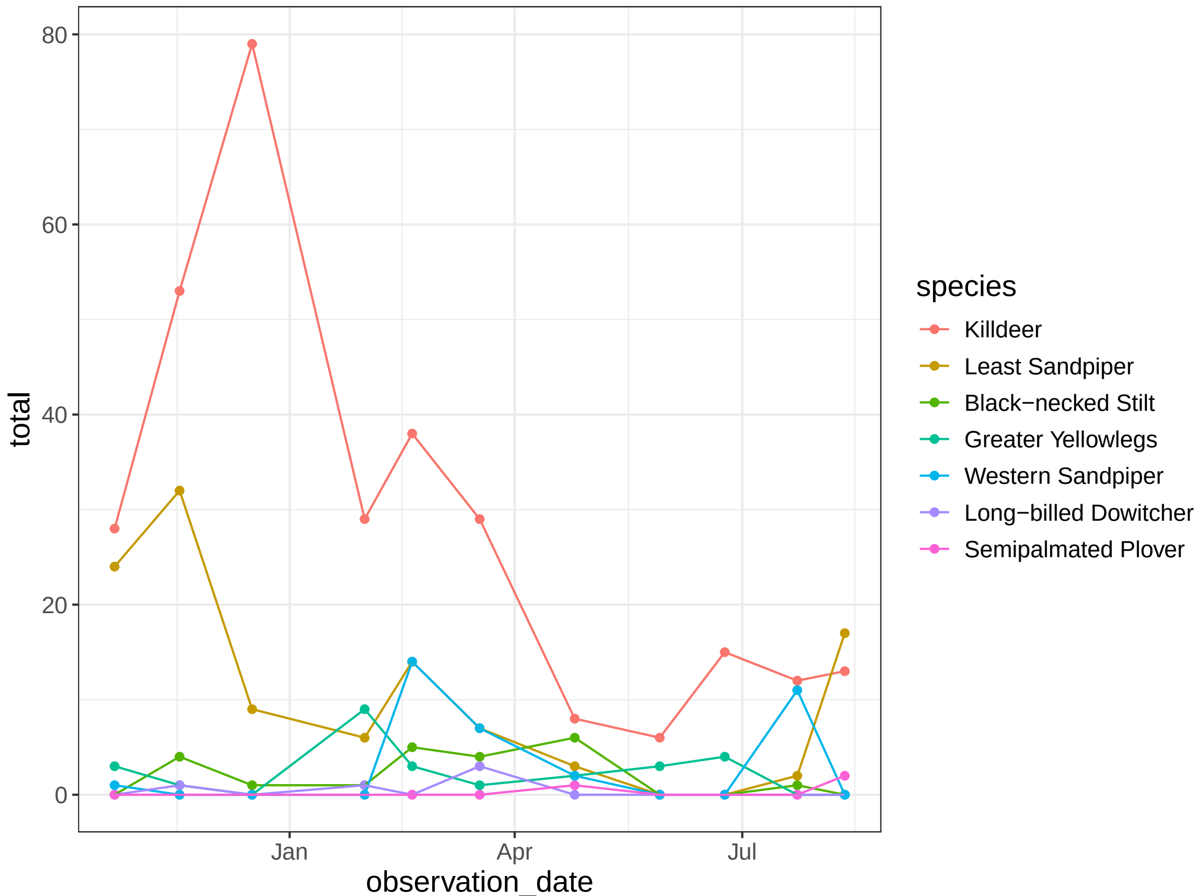
Table of contents

1. Timeseries plots

a. Basic plot

Issues:

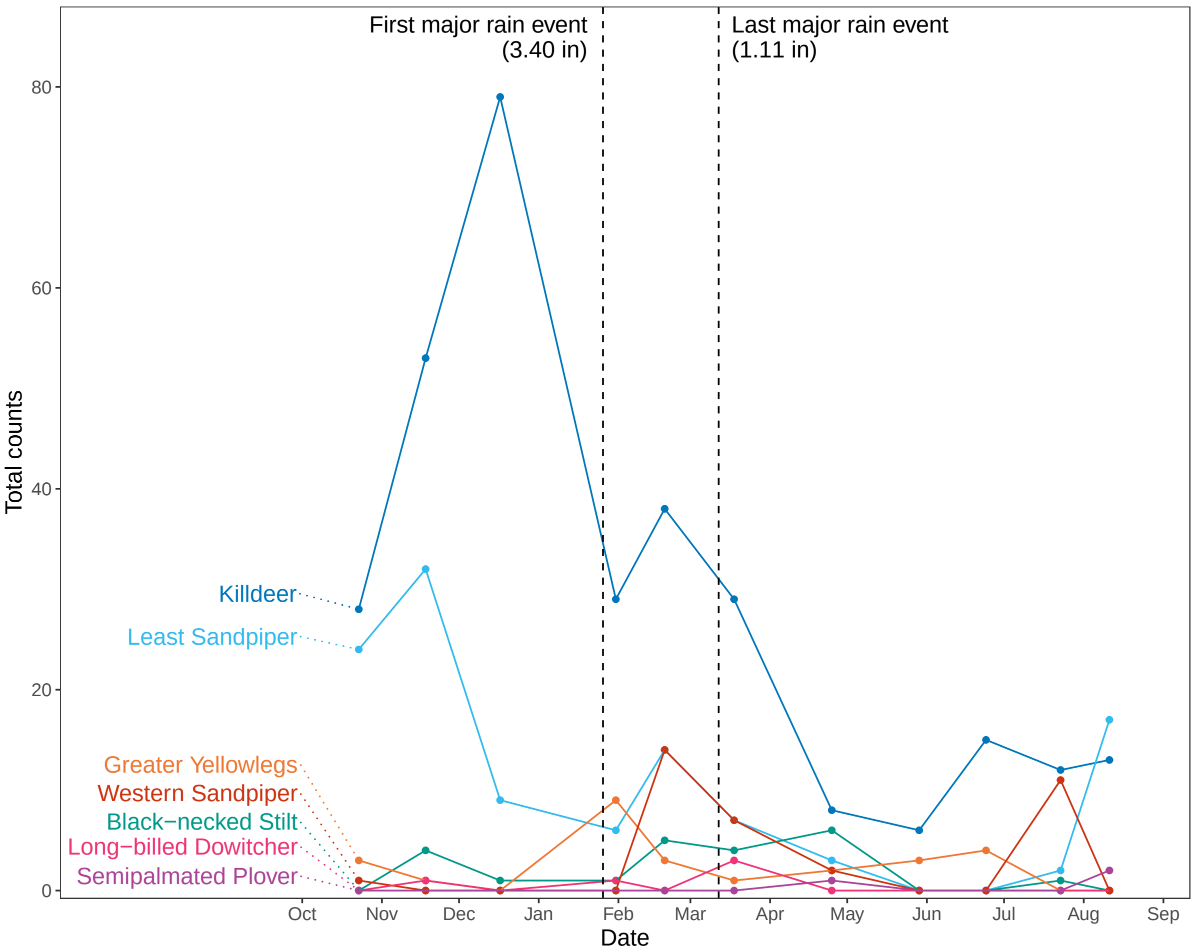
- legend takes up space
- hard to distinguish trends between species



b. Direct labelling

Improvements:

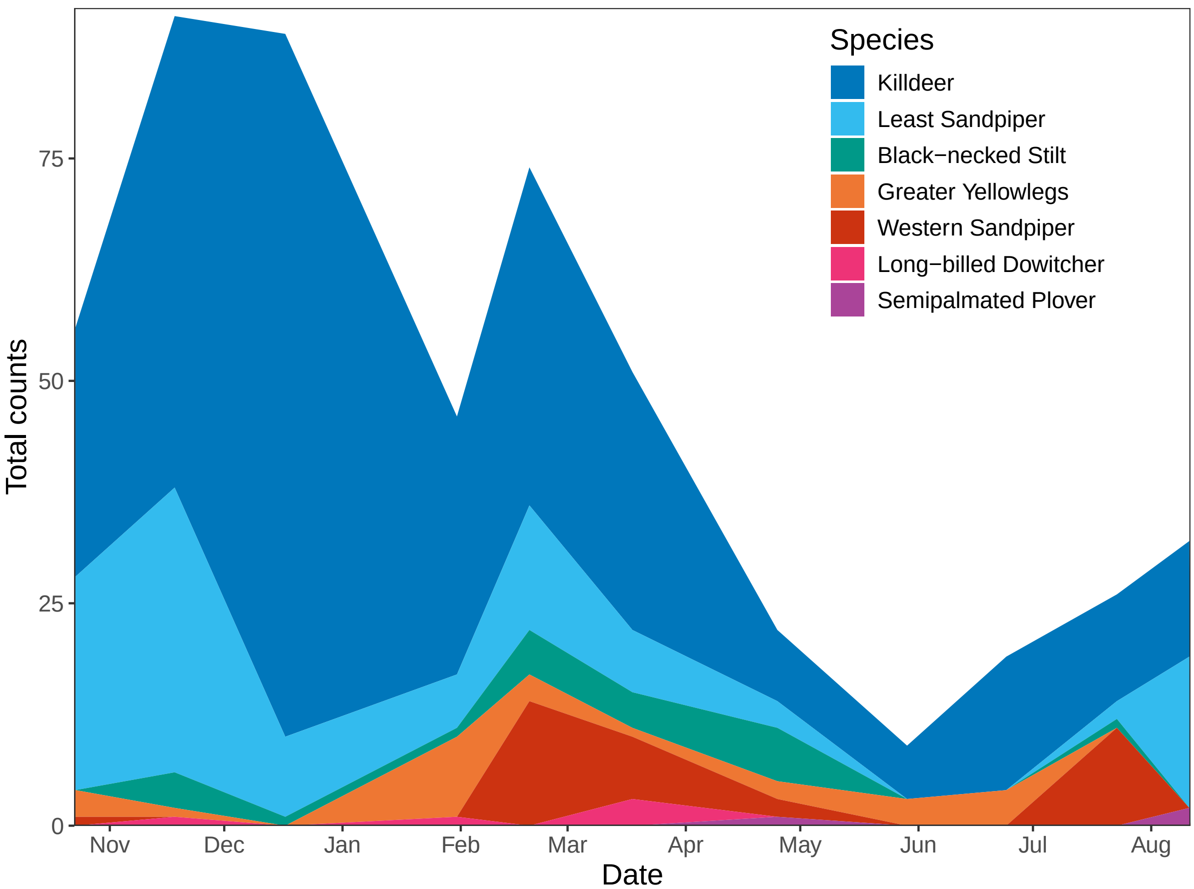
- directly labelled species (getting rid of legend)
- highlighting major events in time



c. Filled areas

Improvements:

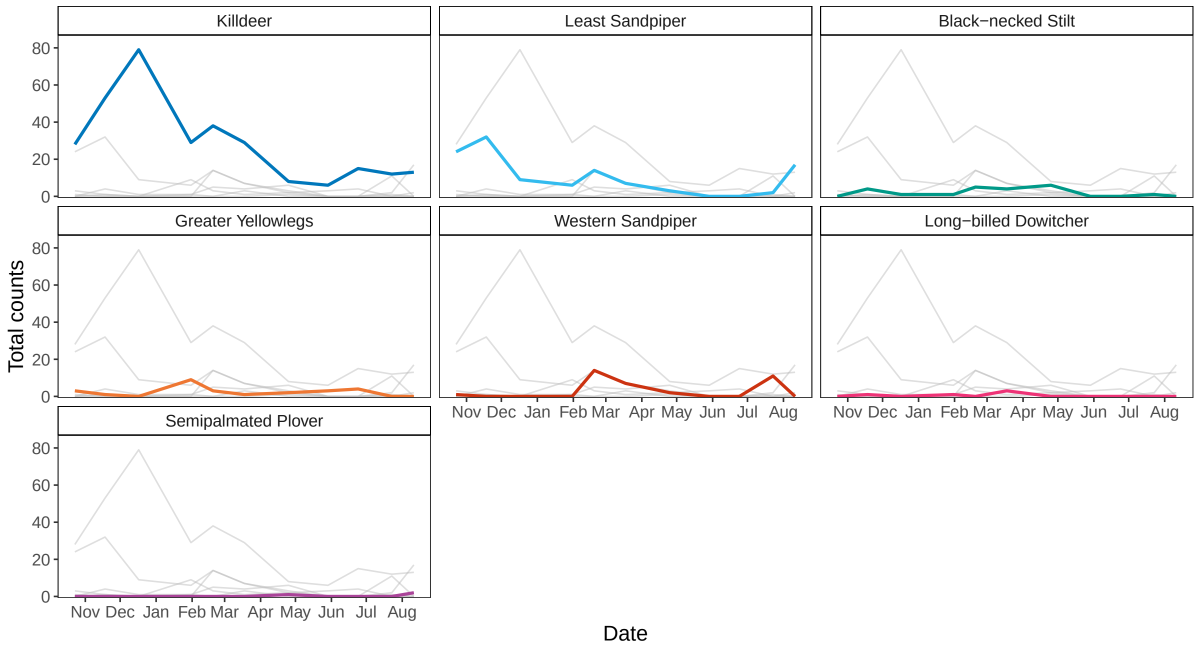
- easier to compare areas than points/lines
- legend inside plot area (takes up less space)



d. Highlighting lines in multiple plots

Improvements:

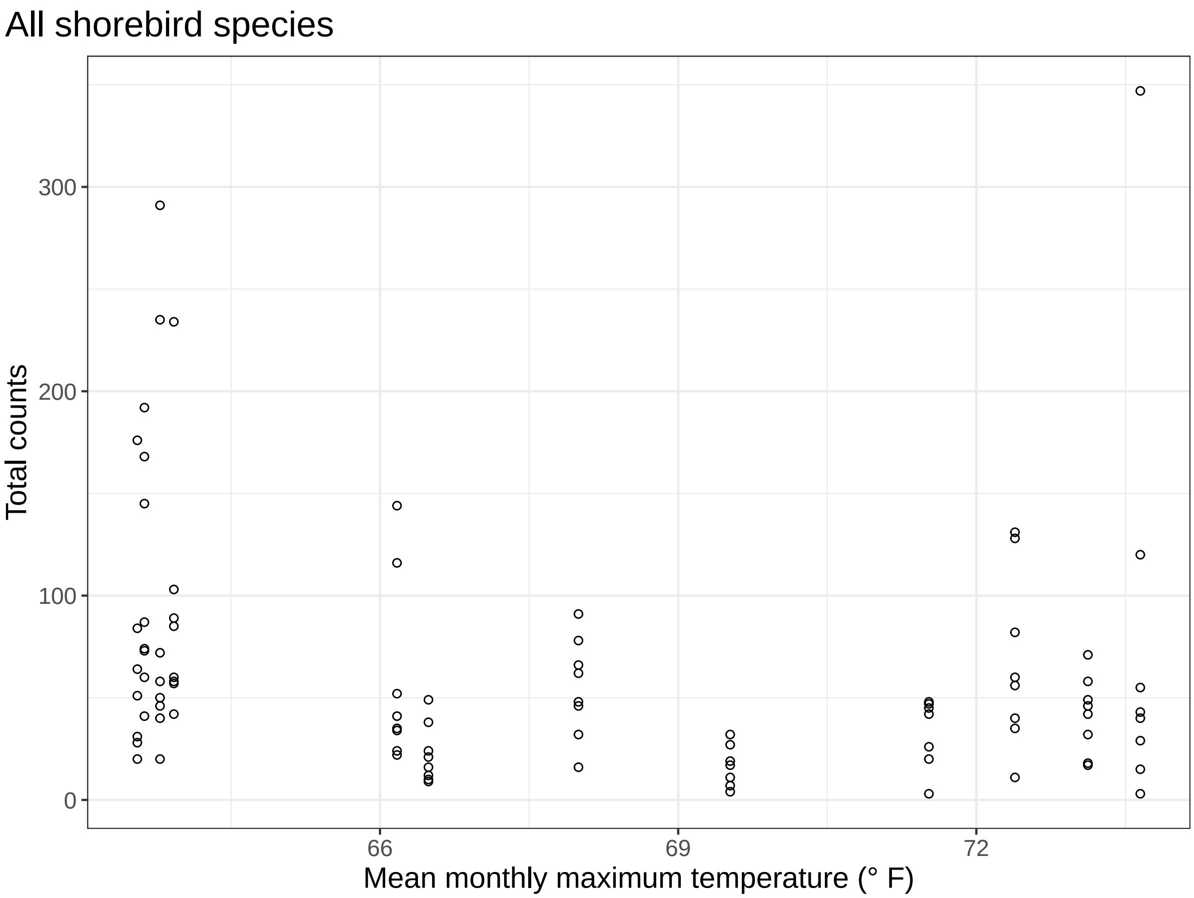
- showing each species individually while keeping other species to compare
- easier to see trends for each species



2. Relationships between variables

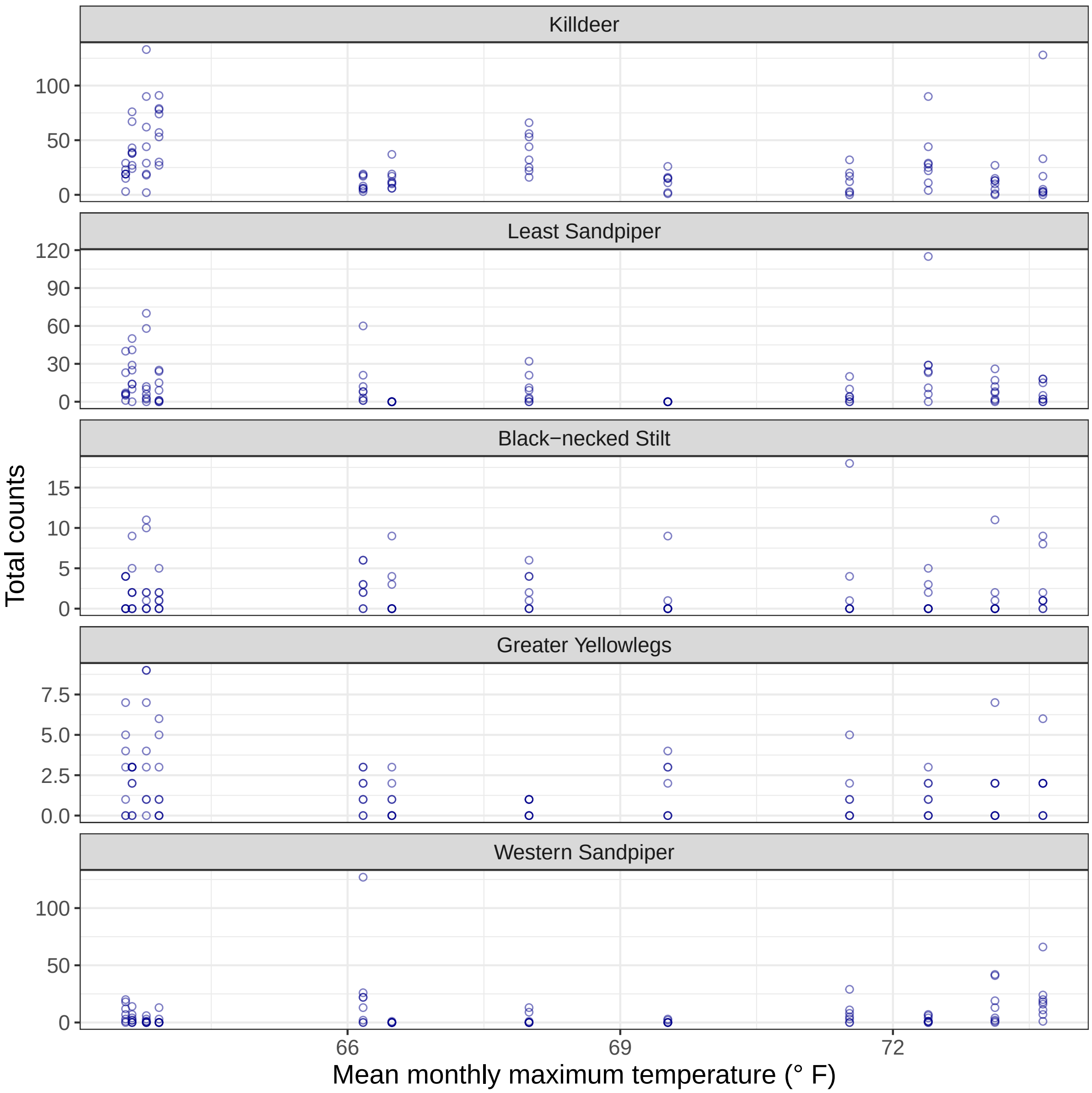
a. Total shorebird counts as a function of temperature

Each point represents a survey. The y-axis represents the total number of shorebirds counted in a survey, and the x-axis represents the mean monthly maximum temperature for the month of that survey.



b. Total species counts as a function of temperature

Each point represents a survey. The y-axis represents the total number counted in a survey for a given species, and the x-axis represents the mean monthly maximum temperature for the month of that survey.



3. Analyses to examine relationships

If you think one variable *predicts* another (i.e. you think you could guess at the total counts for a species if you knew a given maximum temperature), you should use a **linear model.**

If you think the variables are related but one variable doesn't necessarily predict another, you should use **correlation**.

There are two types of correlation: **Pearson correlation** and **Spearman correlation**.